ALICE L WALTON SCHOOL OF MEDICINE

GI Hormones and Intestinal Phase Coordination

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Learning Objectives (5)

After completing this brick, you will be able to:

1. Explain the intestinal phase of digestion and identify the major enteroendocrine cell types, their locations, luminal stimuli for hormone release, and primary target organs.
2. Trace the mechanism by which CCK coordinates pancreatic enzyme secretion, ga[illegible]f gastric emptying in [illegible]
3. Trace the mechanism by which secretin stimulates pancreatic bicarbonate secretion and inhibits gastric acid output in response to luminal acid.
4. Predict how chyme composition and volume regulate gastric emptying rate via duodenal neural and hormonal feedback mechanisms.
5. Predict consequences of impaired duodenal acid neutralization on pancreatic enzyme activity and mucosal integrity.

CASE CONNECTION

TR, a 34-year-old male, presents at an outpatient clinic with bloating, intermittent abdominal pain, and fatty stools (steatorrhea). He reports that symptoms worsen after eating high-fat meals. His primary care physician suspects a problem with digestive coordination in the small intestine.

What GI hormones are implicated in coordinating digestion after a meal? How might a disrupted duodenal environment impact digestion? Consider your answers as you read, and we'll revisit TR at the end of the brick.

GO TO NEXT CONNECTION ↓

Enteroendocrine Cells and the Intestinal Phase of Digestion (LO1)

Digestion is classically divided into three overlapping phases: cephalic, gastric, and intestinal. The intestinal phase begins when chyme—the semi-liquid mixture of partially digested food and gastric secretions—enters the duodenum. This phase is primarily responsible for fine-tuning digestive output, coordinating pancreatic enzyme and bicarbonate secretion, regulating bile delivery, and controlling the rate at which the stomach empties. Unlike the earlier phases, the intestinal phase relies heavily on hormonal signals released by specialized cells lining the small intestine.

The key players in this phase are enteroendocrine cells—a diverse population of hormone-secreting cells scattered throughout the intestinal epithelium. Although they constitute less than 1% of intestinal epithelial cells, they collectively form the largest endocrine organ in the body. These cells sense luminal contents directly through apical membrane receptors and respond by releasing hormones into the bloodstream or via paracrine signaling to nearby cells.

Major Cell Types

Several enteroendocrine cell types are critical during the intestinal phase. Each is named by a letter designation and secretes one or more hormones with distinct physiological roles. The figure and table below summarize the major cell types and their associated hormones.

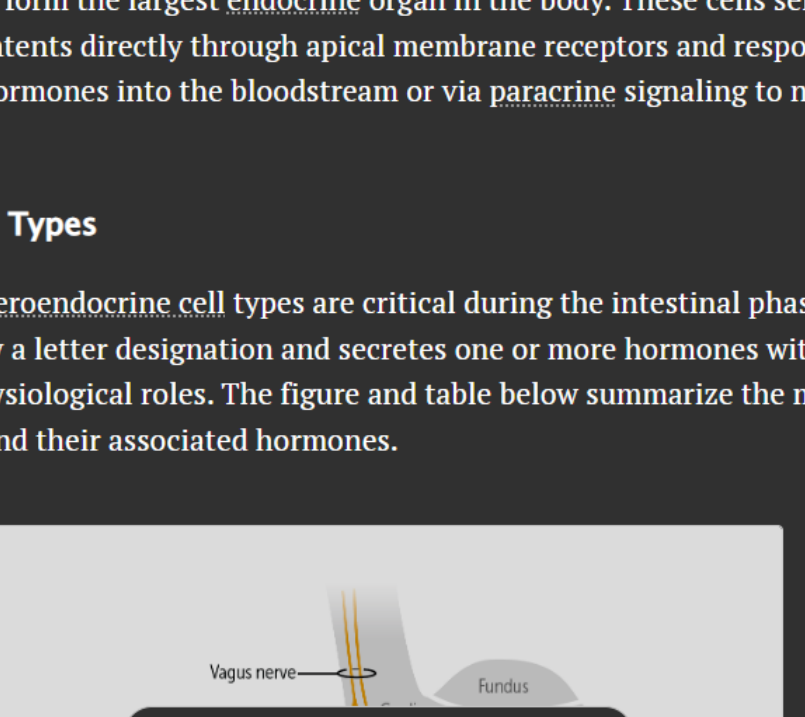

Figure 1: Signaling in the GI system.

Table 1: Key cell types and hormones in the GI system

Cell Type	Location	Luminal Stimulus	Primary Hormone(s)	Key Function
I cells	Duodenum, proximal jejunum	Long-chain fatty acids, peptides, amino acids	Cholecystokinin (CCK)	Enzyme secretion, gallbladder contraction
S cells	Duodenum, proximal jejunum	Luminal acidification (pH <4.5)	Secretin	Bicarbonate secretion, acid inhibition
K cells	Duodenum, jejunum	Glucose, fat	GIP (Glucose-dependent Insulinotropic Peptide)	Insulin release, gastric inhibition
L cells	Ileum, colon	Glucose, fat	GLP-1, PYY	Satiety, motility regulation
G cells	Antrum (predominantly)	[illegible]	[illegible]	Gastric acid stimulation (minor intestinal role)

Target Organs

Each hormone's target organ set is what allows the intestinal phase to coordinate multiple digestive functions in parallel. CCK targets pancreatic acinar cells (enzyme release) and gallbladder smooth muscle (bile expulsion). Secretin acts on pancreatic ductal cells (bicarbonate) and stomach parietal cells (acid inhibition). GIP and GLP-1 target pancreatic β cells to potentiate insulin release, connecting the intestinal phase to postprandial glucose homeostasis. All EECs share a common sensing mechanism: apical chemosensory GPCRs detect specific luminal chemistries and trigger vesicular hormone release into the bloodstream (endocrine) or nearby cells (paracrine).

ACTIVE LEARNING QUESTION

Match the major enteroendocrine cell types and the primary hormone each secretes during the intestinal phase.

GIP · CCK · Gastrin · GLP/PYY · Secretin

I cells	
S cells	
K cells	
L cells	
G cells	

Explanation (requires correct answer)

CCK: Coordinating Pancreatic Secretion, Bile Delivery, and Gastric Emptying (LO2)

Cholecystokinin (CCK) is one of the most important hormones of the intestinal phase. Released from I cells in the duodenum and proximal jejunum, CCK orchestrates a multi-organ response designed to optimize the digestion and absorption of fats and proteins. Its actions span the pancreas, gallbladder, and stomach, making it a central coordinator of postprandial digestion.

CCK Release Mechanism

CCK secretion is triggered primarily by the presence of long-chain fatty acids and partially digested proteins (peptides and amino acids) in the duodenal lumen. These nutrients are detected by apical receptors on I cells, initiating intracellular signaling cascades that lead to CCK exocytosis into the bloodstream. The magnitude of CCK release is proportional to the fat and protein content of the meal—a high-fat, high-protein meal produces a robust CCK response, while a carbohydrate-rich meal elicits minimal CCK secretion.

I cell CCK secretion is also amplified by CCK-releasing factor (CCK-RF), a luminal peptide secreted by duodenal enterocytes. CCK-RF directly stimulates I cells, sustaining CCK release as long as undigested nutrients remain in the lumen. Pancreatic proteases normally degrade CCK-RF; during a meal, these proteases are diverted to food digestion, allowing CCK-RF to persist and maintain the secretory signal. As digestion nears completion and free protease activity rises, CCK-RF is degraded, terminating the stimulus. This protease-dependent mechanism links the duration of CCK secretion to the progress of intraluminal digestion.

Pancreatic and Gallbladder Effects

CCK exerts its digestive effects through two primary target organs: the exocrine pancreas and the gallbladder. In the pancreas, CCK binds to **CCK-A receptors** on acinar cells, stimulating the secretion of digestive enzymes and zymogens (established in the Exocrine Functions brick) into the pancreatic duct for delivery to the duodenum.

Simultaneously, CCK acts on smooth muscle cells of the gallbladder, causing contraction and expulsion of bile into the common bile duct and subsequently into the duodenum via the sphincter of Oddi. Bile salts are essential for emulsifying dietary fats, dramatically increasing the surface area available for lipase activity. CCK also relaxes the sphincter of Oddi, facilitating the coordinated delivery of both bile and pancreatic juice into the duodenal lumen at the same time.

Inhibition of Gastric Emptying

Beyond its effects on the pancreas and gallbladder, CCK plays a critical role in slowing gastric empt[illegible]e pyloric region and via vagal afferent pathways, CCK increases pyloric sphincter tone and reduces antral contractions, effectively slowing the rate at which chyme is delivered to the duodenum. This is physiologically important: it prevents the duodenum from being overwhelmed with more fat and protein than it can process at once.

This inhibitory effect on gastric emptying also contributes to satiety. CCK signals via vagal afferents to the brainstem, reducing appetite and promoting a sense of fullness after a meal.

CASE CONNECTION

Clinically, impaired CCK signaling, as seen in conditions affecting duodenal I cells, can lead to inadequate enzyme secretion, poor fat digestion, and steatorrhea, symptoms that closely mirror those seen in patients like TR.

GO TO NEXT CONNECTION ↓

ACTIVE LEARNING QUESTION

A patient with a duodenal tumor destroying I cells presents with steatorrhea. Which mechanism best explains this finding?

- A. ○ Increased gastric acid output
- B. ○ Reduced pancreatic lipase secretion
- C. ○ Bile d[illegible]
- D. ○ Decreased secretin release
- E. ○ Accelerated gastric emptying

Explanation (requires correct answer)

Secretin: Bicarbonate Secretion and Gastric Acid Inhibition (LO3)

While CCK coordinates enzyme delivery, secretin ensures the duodenal environment is chemically suitable for those enzymes to function. Pancreatic digestive enzymes—particularly proteases and lipase—operate optimally at a near-neutral pH. When acidic chyme enters the duodenum, secretin is released to neutralize it, protecting both the enzymes and the duodenal mucosa.

Secretin Release

Secretin is released from S cells in the duodenum and proximal jejunum when luminal pH drops below approximately 4.5. This acidification occurs as gastric acid-laden chyme enters the duodenum. The lower the pH, the greater the secretin release—a graded response that matches the degree of acid neutralization needed. Fatty acids can also potentiate secretin release, though acid is the dominant stimulus.

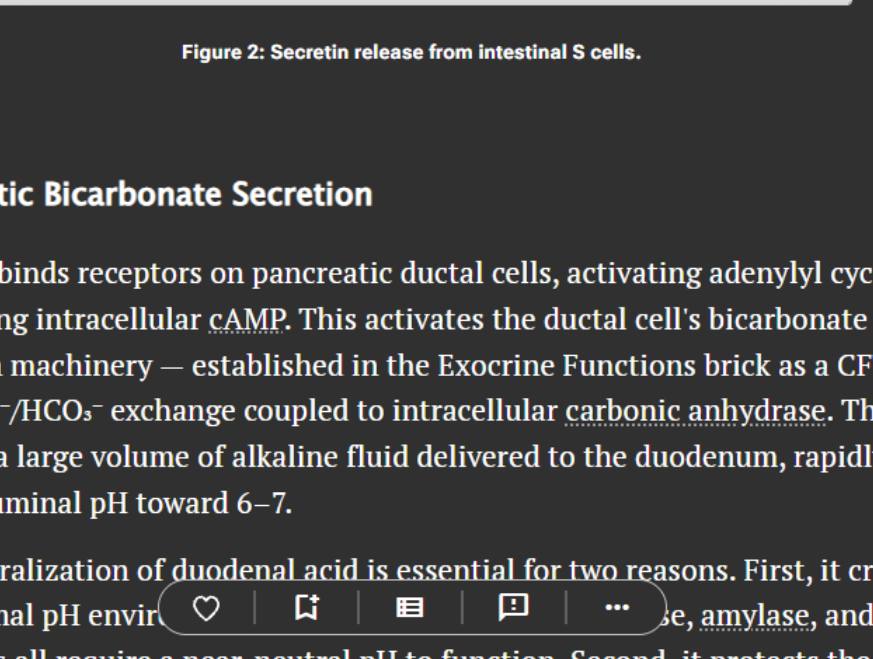

Figure 2: Secretin release from intestinal S cells.

Pancreatic Bicarbonate Secretion

Secretin binds receptors on pancreatic ductal cells, activating adenylyl cyclase and raising intracellular cAMP. This activates the ductal cell's bicarbonate secretion machinery — established in the Exocrine Functions brick as a CFTR-driven Cl⁻/HCO₃⁻ exchange coupled to intracellular carbonic anhydrase. The result is a large volume of alkaline fluid delivered to the duodenum, rapidly raising luminal pH toward 6–7.

The neutralization of duodenal acid is essential for two reasons. First, it creates the optimal pH envir[illegible]e, amylase, and proteases all require a near-neutral pH to function. Second, it protects the duodenal mucosa from acid-induced injury. Secretin also stimulates bile secretion from the liver (choleresis), adding to the buffering capacity of the duodenal contents.

CLINICAL CORRELATION

Cystic Fibrosis and Pancreatic Digestive Failure

Cystic fibrosis (**CFTR** mutations, most commonly **ΔF508**) disrupts the secretin-driven bicarbonate axis at its core. Even when secretin release from S cells is intact, defective CFTR prevents ductal Cl⁻ and HCO₃⁻ secretion, so the alkaline pancreatic fluid never reaches the duodenum. Persistent duodenal acidity inactivates lipase and produces steatorrhea, while thick ductal secretions progressively destroy exocrine pancreatic tissue. Management combines pancreatic enzyme replacement with acid suppression (to preserve residual enzyme activity in the still-acidic duodenum). **CFTR modulators** (e.g., ivacaftor, elexacaftor) restore partial channel function in responsive mutations.

Inhibition of Gastric Acid

In addition to stimulating bicarbonate secretion, secretin inhibits gastric acid production. It acts on parietal cells of the stomach to reduce acid output and also suppresses gastrin release from G cells, thereby dampening the gastric phase of acid secretion. This dual action—neutralizing existing acid in the duodenum while reducing further acid production in the stomach—creates a coordinated brake on[illegible]

Secretin also inhibits gastric motility, contributing to slower gastric emptying in concert with CCK. Together, these hormones ensure that the duodenum receives chyme at a rate it can neutralize and process, rather than being flooded with acid and nutrients simultaneously.

What are secretin's two main target organs and its key actions at each?

Duodenal Feedback Regulation of Gastric Emptying (LO4)

Gastric emptying is not a passive process—it is tightly regulated by feedback from the duodenum to ensure that the small intestine is never overwhelmed. The rate of emptying is continuously adjusted based on the chemical composition, osmolarity, and volume of chyme already present in the duodenum. This regulation involves both neural reflexes and hormonal signals working in concert.

Role of Chyme Composition

The duodenum is exquisitely sensitive to the macronutrient composition of chyme. Fat is the most potent inhibitor of gastric emptying—even small amounts of fat in the duodenum trigger a significant slowing of gastric motility. Protein has a moderate inhibitory effect, while carbohydrates have the least. Hyperosmolar solutions (high sugar or salt concentrations) also slow emptying, as do highly acidic conditions (low pH). This hierarchy of inhibition reflects the digestive demands of each macronutrient: fats require the most preparation (emulsification, enzyme activity) and therefore the most time.

Duodenal Neural Pathways

Neural regulation of gastric emptying occurs primarily through the **enterogastric reflex**. Mechanoreceptors and chemoreceptors in the duodenal wall detect distension, low pH, hyperosmolarity, or nutrient load, and signal the ENS through inhibitory interneurons. The result is reduced excitatory input to the gastric antrum (slowing peristalsis) and increased pyloric sphincter tone. This reflex operates within seconds — an immediate neural brake that acts before hormonal signals. Vagal pathways carry additional inhibitory signals from the brainstem in response to CCK and other hormones, reinforcing the neural brake.

Hormonal Feedback

Hormonal feedback on gastric emptying involves several intestinal hormones acting in parallel. CCK slows emptying by increasing pyloric tone and activating vagal inhi[illegible]ric motility by inhibiting antral contractions. GIP and GLP-1 also contribute to gastric inhibition, particularly in response to glucose and fat. Peptide YY (PYY), released from L cells in the ileum, produces the "ileal brake"—a powerful inhibition of gastric emptying when nutrients reach the distal small intestine.

The combined effect of these hormonal signals is a finely tuned, meal-responsive system. As the duodenum processes chyme, inhibitory signals diminish, allowing gastric emptying to resume. This dynamic regulation prevents both premature delivery of unprocessed nutrients and unnecessary delays in digestion.

ACTIVE LEARNING QUESTION

A high-fat meal most potently slows gastric emptying. Which combination of mechanisms best explains this?

- A. ○ Secretin release alone
- B. ○ Increased gastrin secretion
- C. ○ CCK release and enterogastric reflex
- D. ○ GLP-1 release alone
- E. ○ Increased antral contractions

Explanation (re[illegible]

Consequences of Impaired Duodenal Acid Neutralization (LO5)

The duodenum operates within a narrow pH window. When bicarbonate secretion is impaired—whether due to pancreatic insufficiency, reduced secretin signaling, or conditions like cystic fibrosis—the consequences extend beyond simple discomfort. Persistently acidic conditions in the duodenum disrupt enzyme function, impair nutrient absorption, and threaten the integrity of the mucosal lining.

Pancreatic Enzyme Activity

Pancreatic enzymes are highly pH-sensitive. Lipase, the enzyme responsible for fat digestion, is irreversibly denatured at pH below 4 and loses significant activity below pH 5. Proteases such as trypsin and chymotrypsin also require a near-neutral pH for optimal activity. When the duodenal pH remains low due to inadequate bicarbonate secretion, these enzymes are inactivated before they can complete digestion. The result is malabsorption of fats, proteins, and fat-soluble vitamins (A, D, E, K).

Clinically, impaired fat digestion manifests as steatorrhea—bulky, greasy, foul-smelling stools with high fat content. Steatorrhea is a hallmark of exocrine pancreatic insufficiency and can also result from any condition in which the duodenal acid load exceeds buffering capacity.

Duodenal Mucosal[illegible]

The duodenal mucosa shares the general defensive architecture of gastric mucosa (mucus-bicarbonate layer, tight junctions, and rapid epithelial turnover) but with a critical addition: **Brunner's glands** in the submucosa of the proximal duodenum secrete alkaline mucus in response to acid and secretin stimulation. When acid neutralization fails, these defenses are overwhelmed. Prolonged low-pH exposure disrupts tight junctions, damages surface epithelial cells, and triggers an inflammatory response that can progress to duodenal erosions and ulcers.

Brunner's glands, located in the submucosa of the proximal duodenum, secrete alkaline mucus in response to acid and secretin stimulation. In conditions of chronic acid exposure, these glands can become hyperplastic as a compensatory response. However, if the acid burden exceeds their capacity, mucosal injury ensues. *H. pylori* infection compounds this vulnerability by impairing defenses and increasing acid exposure, particularly in the duodenal bulb.

ACTIVE LEARNING QUESTION

A patient with chronic pancreatitis has reduced bicarbonate secretion. Which outcome is most likely?

- A. ○ Lipase inactivation and steatorrhea
- B. ○ Increased CCK secretion
- C. ○ Accelerated gastric emptying
- D. ○ Increased bile secretion
- E. ○ Reduced mucosal permeability

Explanation (requires correct answer)

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to TR, what GI hormones are implicated in coordinating digestion after a meal? How might a disrupted duodenal environment impact digestion?

TR's postprandial bloating, abdominal pain, and steatorrhea worsened by high-fat meals point to a breakdown in the CCK–secretin axis that coordinates the intestinal phase. Two categories of failure produce this pattern.

Failure of enzyme delivery. If pancreatic enzyme secretion is inadequate — through disrupted CCK signaling or, more commonly, exocrine pancreatic insufficiency itself — dietary triglycerides cannot be hydrolyzed and steatorrhea results. Cystic fibrosis and chronic pancreatitis are the leading causes; TR's age and chronic postprandial presentation warrant workup with fecal elastase and, if CF is suspected, sweat chloride testing or CFTR genotyping.

Failure of duodenal alkalinization. If bicarbonate delivery is inadequate — through CFTR dysfunction in CF or through acid supply exceeding neutralizing capacity in Zollinger-Ellison syndrome — duodenal pH remains low and pancreatic lipase is inactivated even when secreted. ZES adds recurrent peptic ulcers to the picture and is diagnosed by elevated fasting serum gastrin with a positive secretinstimulation [illegible]

Management is diagnosis-specific: pancreatic enzyme replacement therapy (PERT) for exocrine insufficiency, acid suppression to preserve residual lipase activity in the still-acidic duodenum, and MCT supplementation to bypass the lipase-dependent absorption pathway when nutritional status is at risk. For TR, narrowing the differential between CF, chronic pancreatitis, and ZES defines the treatment path.

Summary

- The intestinal phase of digestion is driven by enteroendocrine cells (I, S, K, L cells) in the duodenum and jejunum that sense luminal nutrients and acid, releasing hormones to coordinate pancreatic, biliary, and gastric responses.
- CCK, released from I cells in response to luminal fat and protein, stimulates pancreatic enzyme secretion, gallbladder contraction, and slows gastric emptying—ensuring coordinated delivery of bile and enzymes to the duodenum.
- Secretin, released from S cells when duodenal pH falls below 4.5, stimulates pancreatic ductal bicarbonate secretion via cAMP/CFTR and inhibits gastric acid output, neutralizing the duodenal environment for optimal enzyme activity.
- Gastric emptying is regulated by duodenal feedback: fat, acid, hyperosmolarity, and distension activate the enterogastric reflex and trigger CCK, secretin, GIP, and PYY release, all of which slow gastric emptying to match duodenal processing capacity.
- Impaired duode[illegible]bicarbonate secretion or excess acid load—inactivates pH-sensitive pancreatic enzymes (especially lipase), causing fat malabsorption and steatorrhea.
- Persistent duodenal acidification overwhelms mucosal defenses (mucus-bicarbonate layer, Brunner's glands, tight junctions), leading to epithelial injury, erosions, and ulcer formation.
- Clinical conditions such as exocrine pancreatic insufficiency and Zollinger-Ellison syndrome illustrate the consequences of disrupted intestinal phase coordination, underscoring the importance of hormonal balance in normal digestion.

Review Questions

1. Which enteroendocrine cell primarily releases CCK in response to fat and protein?

- A. ○ I cells
- B. ○ S cells
- C. ○ K cells
- D. ○ L cells
- E. ○ G cells

Explanation (requires correct answer)

2. After fatty acids and amino acids enter the duodenum, which hormone coordinates pancreatic enzyme secretion, gallbladder contraction,and slowed gastric emptying?

- A. ○ Secretin
- B. ○ Gastrin
- C. ○ CCK
- D. ○ GLP-1
- E. ○ Motilin

Explanation (requires correct answer)

3. Secretin released in response to duodenal acid primarily acts on which organs?

- A. ○ Pancreatic enzyme secretion and gallbladder contraction
- B. ○ Pancreatic bicarbonate secretion and gastric acid inhibition
- C. ○ Pancreatic insulin release and hepatic glucose uptake
- D. ○ Gastric acid stimulation and antral contraction
- E. ○ Intestinal[illegible]

Explanation (requires correct answer)

4. Which mechanism slows gastric emptying when fat-rich chyme enters the duodenum?

- A. ○ Increased antral contractions
- B. ○ Pyloric sphincter relaxation
- C. ○ Enterogastric reflex
- D. ○ Migrating motor complex
- E. ○ Receptive relaxation

Explanation (requires correct answer)

5. What is the main consequence of impaired duodenal acid neutralization?

- A. ○ Lipase inactivation and steatorrhea
- B. ○ Increased CCK secretion
- C. ○ Accelerated gastric emptying
- D. ○ Increased bile secretion
- E. ○ Reduced mucosal permeability

Explanation (requires correct answer)

How would you rate this Brick?

👍 👎

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

- 01 ☐ Explain the intestinal phase of digestion and identify the major enteroendocrine cell types, their locations, luminal stimuli for hormone release, and primary target organs.
- 02 ☐ Trace the mechanism by which CCK coordinates pancreatic enzyme secretion, gallbladder contraction, and inhibition of gastric emptying in response to luminal fat and protein.
- 03 ☐ Trace the mechanism by which secretin stimulates pancreatic bicarbonate secretion and inhibits gastric acid output in response to luminal acid.
- 04 ☐ Predict [illegible]late gastric emptyi[illegible]eedback mechanisms.

How would you rate this Brick?

👍 👎

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- 05 ☐ Predict consequences of impaired duodenal acid neutralization on pancreatic enzyme activity and mucosal integrity.

Objective Confidence: 0/5